

Dynamic Graph Neural Policies for Multi-Robot Cooperative Navigation: Edge Features, Safety Filtering, and Density Limits

Xuelin Hu

*Liuzhou Railway Vocational
Technical College*

Liuzhou 545000, China
huxl@ltzy.edu.cn

Jingchao Wang*

*School of Computer Science
Zhongyuan University
of Technology*

Zhengzhou 450007, China
static@zut.edu.cn

Xiaoqin Fu

*Liuzhou Railway Vocational
Technical College*

Liuzhou 545000, China
fuxiaoqin@ltzy.edu.cn

Pengming Hu

*School of Computer Science
Zhongyuan University
of Technology*

Zhengzhou 450007, China
hupengming@zut.edu.cn

Youjing Fu

*College of Earth and
Environmental Sciences
Lanzhou University*

Lanzhou 730000, China
fuyj2025@lzu.edu.cn

Simeng Li

*Liuzhou Railway Vocational
Technical College*

Liuzhou 545000, China
lisimeng@ltzy.edu.cn

Abstract—Multi-robot navigation requires coordinated motion and collision avoidance under changing neighbor relations and limited communication. This paper presents a dynamic graph neural policy that represents robots as nodes and local communication links as edges. Five-dimensional relative-motion edge features encode relative position, velocity, and closing speed, while shared message passing produces local velocity commands for variable-size teams. The policy is trained by imitation with separation, goal-progress, and smoothness losses, and an external safety filter corrects approaching robot pairs during execution. Evaluation covers four map families and teams of 5–40 robots, with 30 unseen seeds per multi-scale configuration. Local message passing improves goal-reaching success over an edge-free policy of the same capacity. Safety filtering reduces close encounters and increases minimum inter-robot separation. In the tested 5–10-robot settings, the filtered policies achieve high collision-free success while retaining millisecond-level control time. These results support the use of local graph interaction and safety filtering for cooperative navigation at low to moderate team densities.

Index Terms—multi-robot systems, graph neural networks, cooperative navigation, collision avoidance, dynamic interaction graphs.

I. INTRODUCTION

Multi-robot cooperative navigation requires robots to reach assigned goals without colliding with one another or with static obstacles. As the number of robots increases, joint planning becomes more costly and local interactions become harder to coordinate. These interactions also change as neighbors enter

and leave sensing range. Centralized search must account for the joint state space, whereas independent policies do not explicitly model the relationships among robots.

Graph neural networks (GNNs) represent these relationships through robot nodes and communication edges. Message passing aggregates neighbor information, and parameter sharing allows the same policy to process different team sizes. Rebuilding edges from current positions updates the interaction structure as the robots move.

We examine how local graph communication affects goal-reaching success relative to a policy using only its own node features. We also evaluate how communication radius and safety filtering affect collision risk and control time. Comparisons use the same observations, action limits, and termination rules.

Our contributions are:

- a dynamic graph policy using five-dimensional edge features for normalized relative position, relative velocity, and closing speed, with shared message passing for variable-size robot teams;
- a training and execution scheme combining action imitation, one-step separation, goal progress, and smoothness losses with an optional external pairwise safety filter, whose effect is evaluated separately from the neural policy;
- a reproducible evaluation across four map families, 5–40 robots, and 30 unseen seeds, with classical and neural baselines and graph, radius, and safety ablations to measure the effects of robot density.

This work was supported by the Science and Technology Research Project of Henan Province (grant no. 262102211055) and the Key Scientific Research Project of Colleges and Universities in Henan Province (grant no. 27AQ520018).

*Corresponding author: Jingchao Wang.

II. RELATED WORK

A. Classical and Rule-Based Planning

A* uses heuristic search to find paths on a discretized map [1]. In multi-robot settings, independently planned paths still require coordination to avoid conflicts. Velocity-obstacle methods identify velocities that would lead to collisions under predicted relative motion [2]. Their geometric formulation supports local collision avoidance, with performance depending on the prediction horizon and the assumptions about neighboring agents' motion.

B. Learned Multi-Robot Collision Avoidance

Decentralized deep reinforcement learning maps local observations to motion commands, reducing the need for joint planning [3], [4]. Curriculum learning introduces increasingly difficult constrained passages [5]; physics-informed priors guide policy learning [6]; and semi-supervised methods reduce labeling requirements [7]. Related studies address communication and formation constraints in vessel navigation [8], [9].

C. Graph Representations for Robot Teams

Diao *et al.* encode spatial relations with a GNN for path planning [10]. He *et al.* combine graph attention with hierarchical motion planning to weight neighboring agents [11]. Goarin and Loianno use a GNN for decentralized goal assignment [12], while Yang *et al.* combine a GNN with PPO for multi-UAV planning [13]. A recent survey reviews rule-based and learned coordination methods for multi-robot collision avoidance [14].

This work combines a graph policy for local motion with a separate filter for approaching robot pairs. Graph and filter ablations distinguish the effects of learned interaction from those of action correction. Tests across four map families and five team sizes assess how these effects change with robot density.

III. PROBLEM FORMULATION

Let $\mathcal{V} = \{1, \dots, N\}$ index the robots, with position $p_i \in \mathbb{R}^2$, velocity $v_i \in \mathbb{R}^2$, and goal $g_i \in \mathbb{R}^2$. A directed communication graph is rebuilt at every step from a radius R_c :

$$A_{ij} = \begin{cases} 1, & i \neq j, \|p_i - p_j\| \leq R_c, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Node features are eight-dimensional: goal-relative position, normalized velocity, direction and distance to the nearest obstacle, and a bias term. The policy emits a two-dimensional local velocity u_i with $\|u_i\| \leq v_{\max} = 1.4$ m/s, and positions evolve as $p_i^{t+1} = p_i^t + \Delta t u_i^t$ with $\Delta t = 0.08$ s.

We distinguish safety-distance violations from physical collisions. A *safety violation* occurs when a pair is closer than $d_s = 0.72$ m; a *physical collision* occurs when the distance is less than the sum of the robot radii, 0.56 m. An event is counted when a pair crosses from outside to inside the corresponding threshold; a sustained violation counts as one event. Task success requires the final mean goal error across

all robots to fall below a fixed threshold. Strict collision-free success additionally requires that no physical collision occur during the episode.

Relative position and velocity are normalized by map size and maximum speed, respectively. For a directed edge (j, i) , the five-dimensional feature is

$$e_{ij}^t = \left[\frac{p_i^t - p_j^t}{[W, H]}, \frac{v_i^t - v_j^t}{v_{\max}}, \frac{-((v_i^t - v_j^t) \cdot \hat{r}_{ij}^t)}{v_{\max}} \right], \quad (2)$$

with $\hat{r}_{ij}^t = (p_i^t - p_j^t) / (\|p_i^t - p_j^t\| + \epsilon)$ and $[W, H]$ the map extents. The closing-speed component is positive when the robots approach each other, providing information about relative motion in addition to distance.

IV. METHOD

A. System Overview

Fig. 1 organizes the closed loop from left to right: the robot scene and observed states, structured graph inputs, feature embedding, the shared GNN policy, and velocity output with optional safety filtering. Each robot supplies an eight-dimensional node feature; communication-radius edges carry five-dimensional relative-motion features. The shared encoder and three message-passing layers produce node representations for a shared two-dimensional velocity head. Executed actions update the scene and the graph is rebuilt at the next control step. Candidate actions illustrate motion choices; the learned policy directly regresses velocity rather than receiving a candidate-action list as input.

B. Baselines

The baselines include A* grid tracking, an ORCA-style local repulsion controller, and an RVO-style controller that samples a finite lattice of headings and speeds. The RVO-style controller scores candidate velocities by predicted neighbor clearance, obstacle occupancy, and goal progress. It is implemented for this benchmark and is not a formally verified RVO reference implementation. The independent MLP shares parameters across robots but uses only node features, without the adjacency matrix.

C. Message-Passing Policy

A linear layer first maps node features to a hidden representation. Each layer averages neighbor representations and updates the node from its own state concatenated with the transformed neighborhood message:

$$m_i^{(l)} = \frac{1}{\max(1, d_i)} \sum_j A_{ij} h_j^{(l)}, \quad h_i^{(l+1)} = \sigma(W_l[h_i^{(l)} \parallel \phi_l(m_i^{(l)})]). \quad (3)$$

With edge features the message becomes

$$m_i^{(l)} = \frac{1}{\max(1, d_i)} \sum_j A_{ij} (\phi_l(h_j^{(l)}) + \psi_l(e_{ij})), \quad (4)$$

followed by $h_i^{(l+1)} = \text{ReLU}(W_l[h_i^{(l)} \parallel m_i^{(l)}])$. We use three message-passing layers, 128 hidden units, and a shared action head, with the action bounded by $\hat{u}_i = v_{\max} \tanh(f_\theta(h_i))$.

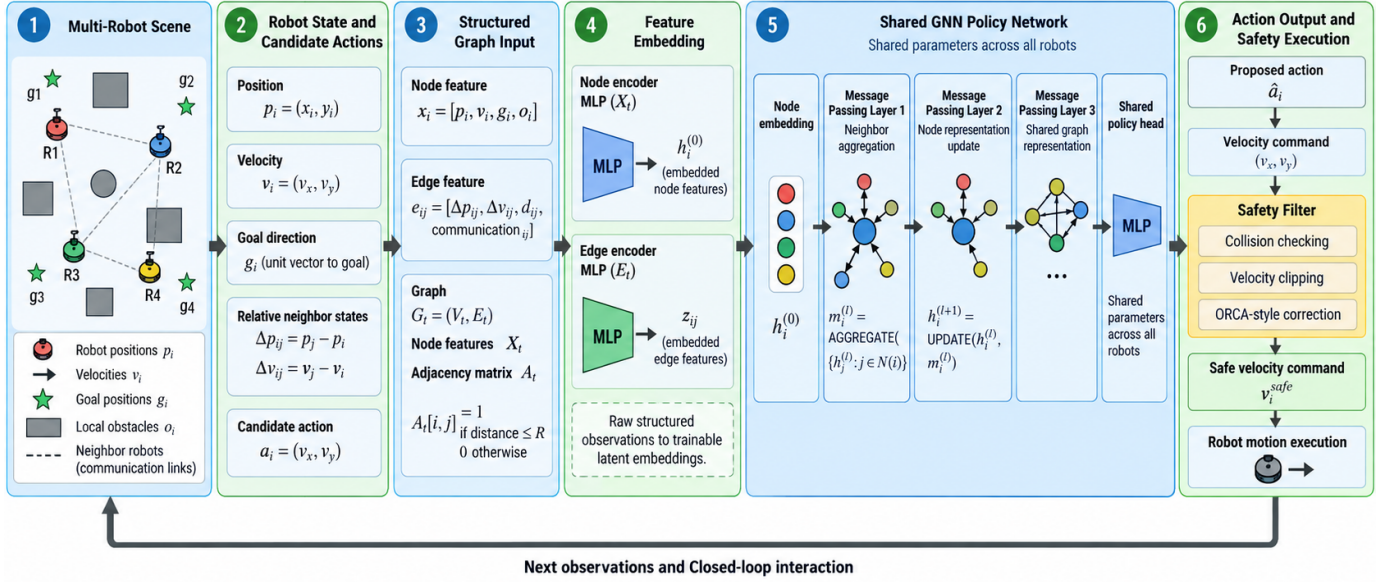


Fig. 1. Multi-robot navigation pipeline: robot scene and states, structured graph inputs, feature embedding, shared GNN inference, and action output with optional safety filtering. Executed motion closes the observation–action loop.

D. Attention Variants

Two attention models are evaluated with the same inputs, outputs, data, and test scenarios. The neighbor-attention GNN replaces mean aggregation with multi-head attention over queries, keys, and values, masked by the local adjacency. The graph transformer uses adjacency-masked multi-head self-attention, residual connections, and feed-forward blocks. Both models use 128 hidden units, four heads, and a shared policy head.

E. Training Objective

Pure action imitation reproduces the expert’s local behavior but does not guarantee a safe closed loop, so we train on a weighted sum of four terms:

$$\mathcal{L} = \lambda_a \mathcal{L}_{act} + \lambda_c \mathcal{L}_{col} + \lambda_p \mathcal{L}_{prog} + \lambda_s \mathcal{L}_{smooth}, \quad (5)$$

where $\mathcal{L}_{act} = \frac{1}{|V|} \sum_i \text{Huber}(u_i, u_i^*)$ is the imitation term. Writing the one-step predicted pair distance as $d'_{ij} = \|p_i + \Delta t u_i - p_j - \Delta t u_j\|$, $\mathcal{L}_{col} = \frac{1}{N(N-1)} \sum_{i \neq j} [d_s - d'_{ij}]_+^2$ penalizes predicted distances below the safety threshold. The remaining terms are

$$\begin{aligned} \mathcal{L}_{prog} &= \frac{1}{N} \sum_i [\|g_i - p'_i\| - \|g_i - p_i\|]_+, \\ \mathcal{L}_{smooth} &= \frac{1}{N} \sum_i \|u_i^t - u_i^{t-1}\|_2^2, \end{aligned} \quad (6)$$

with $[x]_+ = \max(x, 0)$. Padding nodes are masked by $q_i \in \{0, 1\}$, so every node loss becomes $\sum_i q_i \ell_i / \max(1, \sum_i q_i)$ and a graph padded to 40 nodes trains on exactly the real robots in it.

Multi-scale training samples datasets with 5–40 robots from all four map families and pads each graph to 40 nodes. The progress term penalizes motion away from the goal, while the smoothness term limits changes between successive actions.

F. Safety Filter

After the policy emits its action, the filter inspects every close pair and applies a symmetric correction before clipping. Let $n_{ij} = \delta_{ij} / (\|\delta_{ij}\| + \epsilon)$ with $\delta_{ij} = p_i - p_j$. If $\|\delta_{ij}\| < d_s + \tau$ and $(u_i - u_j) \cdot n_{ij} < 0$, then

$$\begin{aligned} \tilde{u}_i &= u_i + \frac{\kappa_{ij}}{2} n_{ij}, \quad \tilde{u}_j = u_j - \frac{\kappa_{ij}}{2} n_{ij}, \\ \kappa_{ij} &= -(u_i - u_j) \cdot n_{ij} + [d_s + \tau - \|\delta_{ij}\|]_+, \end{aligned} \quad (7)$$

after which each $\tilde{u}_i$ is clipped to $v_{\max}$. The filter operates only during inference; Section VI evaluates its effect separately from that of the learned policy.

V. EXPERIMENTAL SETUP

Experiments span four map families—*crossing*, *open*, *random*, and *narrow*—and 5, 10, 20, 30, and 40 robots. Development data uses fixed seeds with 20 episodes of at most 160 steps each. The main evaluation covers $4 \times 3 \times 5 \times 10 = 600$ episodes over three robot counts, five methods, and ten fixed test seeds; the extended learned-model comparison covers 480 episodes for four learned policies; and the multi-scale test covers 1800 safety-filtered and 1800 unfiltered episodes over 30 unseen seeds (100–129). All methods share initial states, goals, obstacles, action limits, and termination rules. Data are stored as compressed NPZ archives holding features, adjacency, actions, positions, rewards, obstacles, layout, expert identifier, and step index.

Reported metrics are goal success rate, strict no-physical-collision success rate, safety-violation events, physical collision events, minimum inter-robot separation, path length, final goal error, and mean per-step control time. Means, standard deviations, and 95% confidence intervals are computed over

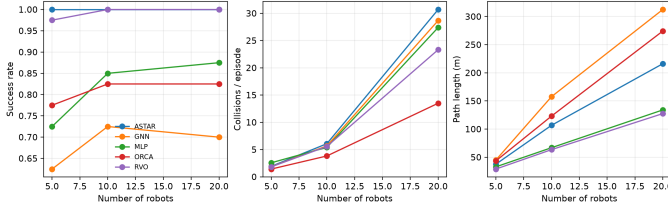


Fig. 2. Comparison of goal success, path length, and control time across methods and team sizes in the 600-episode main evaluation.

episodes, and violations at 0.72 m and physical collisions at 0.56 m are always reported in separate columns. Training used one 24 GB GPU with batch size 128.

VI. RESULTS

A. Main Comparison

Table I summarizes the 600-episode main comparison. A* attains a task success rate of 1.000 at each team size. RVO-style achieves 0.975–1.000 success and the shortest paths, while its mean control time increases from 7.8 ms at 5 robots to 87.4 ms at 20 robots, approximately an 11-fold increase. ORCA-style has the lowest control time and the largest minimum separation at each team size. Among the learned policies in this comparison, the independent MLP achieves the higher success rates (0.725–0.875).

The mean-aggregation GNN reaches success rates of 0.625, 0.725, and 0.700 at 5, 10, and 20 robots. These rates are below those of the MLP, while the GNN has larger minimum inter-robot separation at each team size. The GNN checkpoint was trained only on 10-robot data; its 5- and 20-robot results therefore measure generalization across team sizes. The graph ablation below compares the same network with and without communication edges to isolate the effect of message passing.

B. Attention and Transformer Variants

Table II compares four learned policies over 480 episodes. Goal success is 0.817 for the MLP, 0.683 for the GNN, 0.558 for the graph transformer, and 0.342 for the neighbor-attention GNN. Their strict collision-free success rates are 0.017, 0.008, 0.058, and 0.025, respectively. The graph transformer has the highest strict success rate, with a mean control time of 1.365 ms compared with 0.856 ms for the GNN. Neighbor attention increases mean minimum separation to 0.259 m, but lowers goal success and takes 1.298 ms per step. These unfiltered policies remain insufficient for reliable collision-free navigation.

C. Graph Structure Ablation

The graph ablation compares local, global, and edge-free connectivity (Fig. 3a). At 10 robots, the local graph reaches 0.725 success compared with 0.425 for the edge-free policy of the same capacity, an increase of 30 percentage points. At 5 and 20 robots, the corresponding increases are 12.5 and 27.5 percentage points. Fully connecting the graph does not improve on local communication; at 10 robots, success is

0.625 for the global graph and 0.725 for the local graph. These results support local neighbor exchange for goal reaching, although collisions remain in both graph variants.

D. Safety Filter Ablation

The safety-filter ablation compares raw and corrected GNN actions over 60 episodes (Fig. 3b). At 10 robots, filtering reduces mean safety-violation events from 4.900 to 0.200 and increases mean minimum separation from 0.069 m to 0.794 m, while goal success decreases from 1.000 to 0.900. At 20 robots, violation events decrease from 27.500 to 6.900 and goal success from 0.900 to 0.700; mean minimum separation is 0.067 m. The correction improves spacing but can impede goal progress. Its limited effect at 20 robots motivates further study of safety constraints during training.

E. Communication Radius

Increasing the communication radius adds neighbors to each robot’s graph. At 10 robots, a radius of 2.0 m gives a mean degree of 0.463 and success of 0.200. Radii of 3.4 m and 5.5 m increase mean degree to 2.283 and 4.420, respectively, and both achieve success of 1.000. The global graph has mean degree 9.000 and success 0.900. At 5 robots, success increases from 0.100 at 2.0 m to 1.000 with the global graph. The results indicate that a small radius can restrict information exchange, while additional neighbors beyond an adequate local range do not consistently improve success. Control time ranges from about 0.8 to 1.5 ms across the tested settings, and physical collisions are not eliminated by global connectivity.

F. Multi-Scale Stress Test

Table III reports the 1800-episode safety-filtered test using 30 unseen seeds per configuration. At 5 and 10 robots, GNN strict collision-free success is 0.967 and 0.958, respectively, with no recorded physical collisions. The graph transformer achieves 0.983 and 1.000 under the same conditions. At 20 robots, goal success is 0.758 for the GNN and 0.842 for the transformer, and strict success is 0.708 and 0.767. At 30 and 40 robots, strict success decreases to 0.250 and 0.000 for the GNN, and to 0.242 and 0.017 for the transformer. At 40 robots, mean physical collision events reach 6.883 for the GNN and 9.025 for the transformer, with minimum separation near 0.4 m. GNN mean control time increases from 0.890 ms at 5 robots to 2.066 ms at 40 robots.

Without filtering, the GNN achieves goal success of 0.975–1.000 across team sizes, but strict collision-free success is 0.092 at 5 robots and 0.000 at larger sizes. The unfiltered transformer’s strict success does not exceed 0.117. Filtering therefore improves collision-free arrival, particularly at 5–10 robots, while reducing goal-reaching success. At 30–40 robots, both completion and safety deteriorate in the filtered policies, limiting their use at these densities.

G. Seed Stability

Three independently trained GNN checkpoints give mean success of 0.517 ± 0.388 , 0.767 ± 0.161 , and 0.850 ± 0.115

TABLE I

MAIN COMPARISON OVER FOUR MAP FAMILIES AND TEN FIXED TEST SEEDS (40 EPISODES PER CELL). SUCC: GOAL SUCCESS. COLL: COLLISION EVENTS, COUNTING TRANSITIONS INTO A < 0.72 M SAFETY VIOLATION. SEP: MINIMUM INTER-ROBOT SEPARATION (M). T : MEAN PER-STEP CONTROL TIME (MS).

Method	5 robots				10 robots				20 robots			
	Succ	Coll	Sep	T	Succ	Coll	Sep	T	Succ	Coll	Sep	T
A*	1.000	1.925	0.280	0.512	1.000	6.075	0.098	1.205	1.000	30.700	0.035	2.493
ORCA-style	0.775	1.425	0.396	0.213	0.825	3.825	0.296	0.486	0.825	13.500	0.230	1.303
RVO-style	0.975	1.825	0.294	7.779	1.000	5.575	0.131	24.941	1.000	23.325	0.062	87.373
MLP	0.725	2.575	0.191	0.342	0.850	5.400	0.095	0.483	0.875	27.400	0.038	0.775
GNN	0.625	1.850	0.224	0.683	0.725	5.675	0.149	0.828	0.700	28.675	0.041	1.129

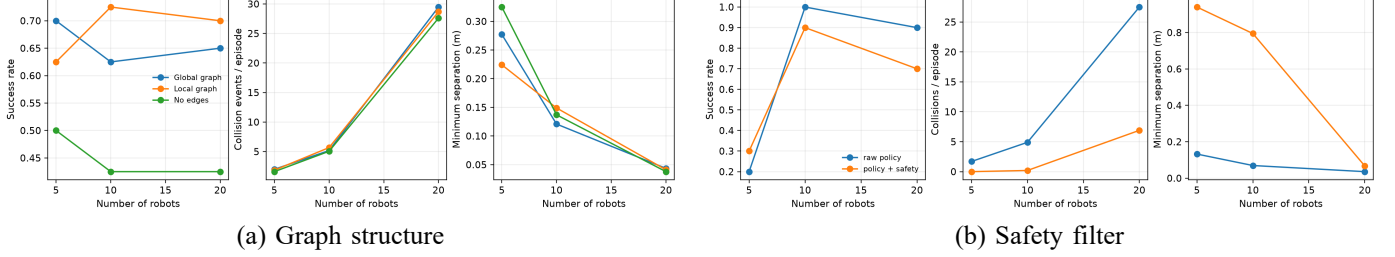


Fig. 3. Ablation results at 5, 10, and 20 robots. (a) Goal success with edge-free, local, and global graphs. (b) Goal success, safety-violation events, and minimum separation before and after safety filtering.

TABLE II

LEARNED-POLICY COMPARISON (480 EPISODES). STRICT SUCCESS REQUIRES GOAL ARRIVAL WITH NO PAIR EVER CLOSER THAN THE 0.56 M PHYSICAL DIAMETER.

Model	Success	Strict succ.	Phys. coll. events	Min. sep. (m)	Time (ms)
MLP	0.817	0.017	9.250	0.108	0.515
GNN	0.683	0.008	8.825	0.138	0.856
Neighbor-attention	0.342	0.025	8.167	0.259	1.298
Graph transformer	0.558	0.058	8.833	0.221	1.365

TABLE III

MULTI-SCALE TEST WITH THE SAFETY FILTER OVER 30 UNSEEN SEEDS (1800 EPISODES), AGGREGATED OVER FOUR MAP FAMILIES AND INDEXED BY ROBOT COUNT. STRICT SUCCESS REQUIRES GOAL ARRIVAL WITH NO PHYSICAL COLLISION.

Model	5	10	20	30	40
<i>Goal success / strict collision-free success</i>					
GNN	0.967/0.967	0.958/0.958	0.758/0.708	0.400/0.250	0.042/0.000
Transformer	0.983/0.983	1.000/1.000	0.842/0.767	0.467/0.242	0.108/0.017
<i>Physical collision events per episode</i>					
GNN	0.000	0.000	0.075	1.158	6.883
Transformer	0.000	0.000	0.217	2.267	9.025
<i>Minimum separation (m) / control time (ms)</i>					
GNN	0.97/0.89	0.95/1.04	0.87/1.37	0.61/1.70	0.41/2.07
Transformer	1.00/1.50	0.94/1.65	0.80/1.99	0.56/2.33	0.34/2.69

at 5, 10, and 20 robots. The standard deviations are computed across checkpoint-level mean success rates. Variation is largest at 5 robots, indicating sensitivity to the training seed in this setting. This analysis evaluates training variability separately from the episode-level averages reported for fixed checkpoints in the preceding comparisons.

VII. LIMITATIONS AND CONCLUSION

This paper presents a dynamic graph policy for multi-robot navigation with relative-motion edge features, a joint imitation

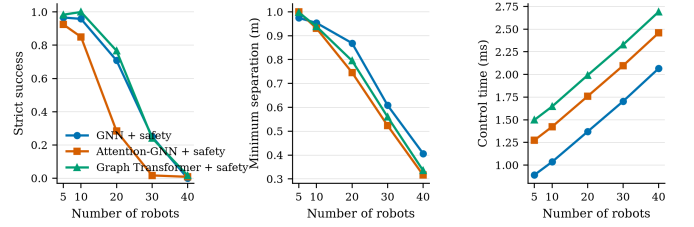


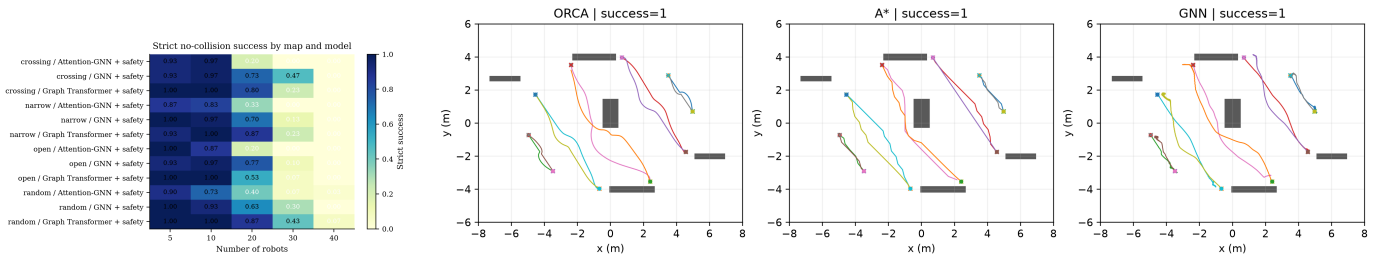
Fig. 4. Strict collision-free success, minimum separation, and mean control time with safety filtering at different team sizes. Collision-free success decreases as the number of robots exceeds 20.

objective, and an external safety filter. Local message passing improves goal success relative to an edge-free policy of the same capacity, while global connectivity offers no consistent additional benefit. Safety filtering increases inter-robot separation and supports high collision-free success at 5–10 robots in the multi-scale test. The GNN maintains millisecond-level control time across the tested team sizes.

The experiments also identify limits. The mean-aggregation GNN does not exceed the independent MLP’s goal success in the main comparison, and strict collision-free success declines at 30–40 robots. The main GNN checkpoint is trained at one team size, and the rule-based expert constrains the behaviors available for imitation. The RVO-style baseline uses finite candidate sampling and should not be equated with a verified RVO implementation. Evaluation is limited to simulation. Further work will examine training with explicit safety constraints, comparison with reference RVO implementations, cross-map generalization, and physical-robot experiments.

REFERENCES

- [1] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Transactions on Systems Science*



(a) Strict success by map and density

(b) Representative trajectories

Fig. 5. (a) Strict collision-free success by map family and population size; *random* and *narrow* layouts are the most demanding at high density. (b) Representative trajectories, with communication edges drawn as transient links between robots within the radius.

and Cybernetics, vol. 4, no. 2, pp. 100–107, 1968.

- [2] P. Fiorini and Z. Shiller, "Motion planning in dynamic environments using velocity obstacles," *The International Journal of Robotics Research*, vol. 17, no. 7, pp. 760–772, 1998.
- [3] P. Long, T. Fanl, X. Liao, W. Liu, H. Zhang, and J. Pan, "Towards optimally decentralized multi-robot collision avoidance via deep reinforcement learning," *IEEE*, 2018, pp. 6252–6259.
- [4] N. Thumiger and M. Deghat, "A multi-agent deep reinforcement learning approach for practical decentralized uav collision avoidance," *IEEE Control Systems Letters*, vol. 6, pp. 2174–2179, 2022.
- [5] M. M. R. Komol, B. Tidd, W. Browne, F. Maire, J. Williams, and D. Howard, "Learning behaviours for decentralised multi-robot collision avoidance in constrained pathways using curriculum reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 10, no. 8, pp. 8538–8545, 2025.
- [6] P. Feng, R. Shi, S. Wang, J. Liang, X. Yu, S. Li, and W. Wu, "Safe and efficient multi-agent collision avoidance with physics-informed reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 9, no. 12, pp. 11 138–11 145, 2024.
- [7] M. Parooei, M. T. Masouleh, and A. Kalhor, "Semi-supervised deep reinforcement learning in decentralized multi-agent collision avoidance and path planning in a complex environment," *IEEE*, 2023.
- [8] Y. Niu, F. Zhu, M. Wei, Y. Du, and P. Zhai, "A multi-ship collision avoidance algorithm using data-driven multi-agent deep reinforcement learning," *Journal of Marine Science and Engineering*, vol. 11, no. 11, p. 2101, 2023.
- [9] Y. Wang and Y. Zhao, "Multiple ships cooperative navigation and collision avoidance using multi-agent reinforcement learning with communication," *Ocean Engineering*, vol. 320, p. 120244, 2025.
- [10] X. Diao, W. Chi, and J. Wang, "Graph neural network based method for robot path planning," *Biomimetic Intelligence and Robotics*, vol. 4, no. 1, p. 100147, 2024.
- [11] X. He, X. Shi, J. Hu, and Y. Wang, "Multi-robot navigation with graph attention neural network and hierarchical motion planning," *Journal of Intelligent & Robotic Systems*, vol. 109, no. 2, 2023.
- [12] M. Goarin and G. Loianno, "Graph neural network for decentralized multi-robot goal assignment," *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4051–4058, 2024.
- [13] Q. Yang, J. Lu, Y. Zhang, and S. Shao, "Multi-uav cooperative path planning via graph neural network and proximal policy optimization," *IEEE Access*, vol. 13, pp. 193 575–193 588, 2025.
- [14] G. Yang, L. An, and C. Zhao, "Collision/obstacle avoidance coordination of multi-robot systems: A survey," *Actuators*, vol. 14, no. 2, p. 85, 2025.